

Question

Write a program to find a fragment that occurs in all strings, where a fragment is 3 consecutive words.

Note: If you are writing this program in JAVA, don't use built-in functions like `split()`, `indexOf()`, `replace()`, `substring()`, etc present in String Class. Do not hard code the output.

Example:

Given three strings like:

S1 = "Every morning I want to do exercise regularly"

S2 = "Every morning I want to do meditation without fail"

S3 = "It is important that I want to be happy always"

Then the :

Common fragment = "I want to"

Answer

```
public class Q1 {

    public static void main(String[] args) {
        String s1="Every morning i want to do exercise regularly";
        String s2="Every morning i want to do meditation";
        String s3="it is important that i want to be happy";

        Q1 start=new Q1();

        //Sucssesfully splited all strings

        String res[]=start.Word_Seperated(s1);
        //for(int i=0;i<res.length;i++) {System.out.print(res[i]+" ");}
        //System.out.println();
        String res1[]=start.Word_Seperated(s2);
        //for(int i=0;i<res1.length;i++) {System.out.print(res1[i]+" ");}
        //System.out.println();
        String res2[]=start.Word_Seperated(s3);
        //for(int i=0;i<res2.length;i++) {System.out.print(res2[i]+" ");}

        //seperated 3 consicutive fragments

        String c1[]=Q1.find3consicutive(res);

        /*/  for(int i=0;i<c1.length;i++) {
            System.out.println(c1[i]);
        }/*/
        String c2[]=Q1.find3consicutive(res1);
        /*/for(int i=0;i<c2.length;i++) {
            System.out.println(c2[i]);
        }/*/
        String c3[]=Q1.find3consicutive(res2);
        /*/
        for(int i=0;i<c3.length;i++) {
            System.out.println(c3[i]);
        }
        /*/

        //finding common fragments
        for(int i=0;i<c1.length;i++) {
```



```

        for(int j=0;j<c2.length;j++) {
            if(c1[i].equalsIgnoreCase(c2[j]))
            {
                for(int k=0;k<c3.length;k++) {
                    if(c1[i].equalsIgnoreCase(c3[k])) {
                        System.out.println("Common fragment is
: "+c1[i]);
                    }
                }
            }
        }
    }

}

public static String[] find3consicutive(String[] res) {
    String cons1[]=new String[res.length-2];
    for(int i=0;i<res.length-2;i++) {
        cons1[i]=res[i]+res[i+1]+res[i+2];
    }
    return cons1;
}

public String[] Word_Seperated(String s1) {

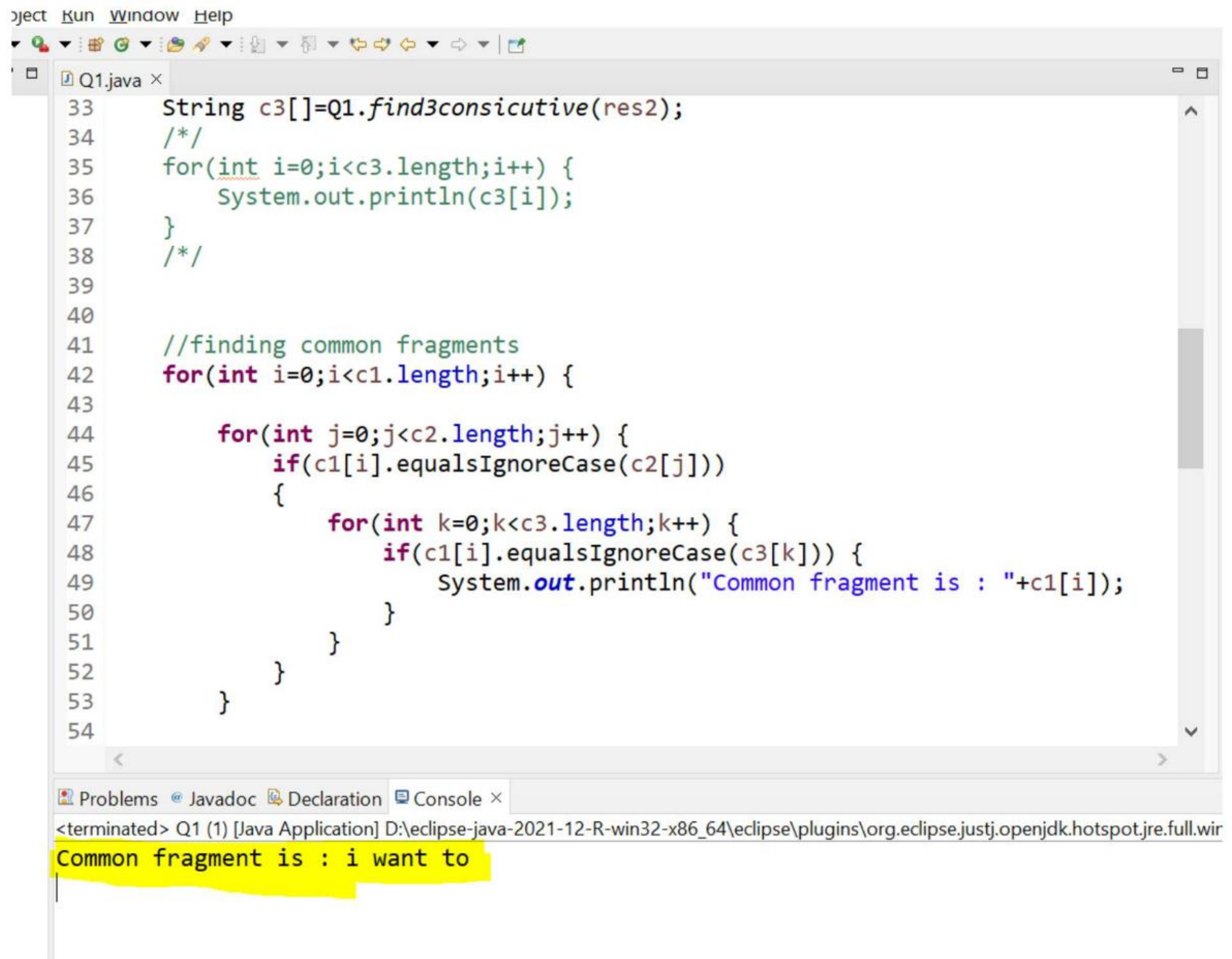
    int Space_count=0;
    for(int i=0;i<s1.length();i++) {
        if(s1.charAt(i)==' ') {
            //System.out.println(Space_count++ +" - "+ i);
            Space_count++;
        }
    }

    String s[]=new String[Space_count+1];
    for(int i=0;i<s.length;i++) {s[i]="";}

    int k=0;
    for(int i=0;i<s1.length();i++) {
        if(s1.charAt(i)!=' ') {
            s[k]=s[k]+s1.charAt(i);
        }else {k++;}
    }
    return s;}}

```


Output proof



The screenshot shows the Eclipse IDE with a Java file named Q1.java. The code defines a method `find3consecutive` that returns an array of strings. It then iterates through the array and prints each element. A comment indicates the next step is to find common fragments. The console output shows the first element of the array, "Common fragment is : i want to".

```
33 String c3[]=Q1.find3consecutive(res2);
34 /**
35 for(int i=0;i<c3.length;i++) {
36     System.out.println(c3[i]);
37 }
38 /**
39
40
41 //finding common fragments
42 for(int i=0;i<c1.length;i++) {
43
44     for(int j=0;j<c2.length;j++) {
45         if(c1[i].equalsIgnoreCase(c2[j]))
46         {
47             for(int k=0;k<c3.length;k++) {
48                 if(c1[i].equalsIgnoreCase(c3[k])) {
49                     System.out.println("Common fragment is : "+c1[i]);
50                 }
51             }
52         }
53     }
54 }
```

<terminated> Q1 (1) [Java Application] D:\eclipse-java-2021-12-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.wir
Common fragment is : i want to